Jingfan Tang

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Education

Shanghai Jiao Tong University Sept 2023 – Aug 2027
B.E. in Electrical & Computer Engineering, GPA: 3.83/4.0 (expected)

Carnegie Mellon University Jan 2026 – May 2026
Department of Electrical & Computer Engineering, College of Engineering

Selected Projects

[1] **HEHA: Bounded Suboptimal Planning for Constrained Routing and Its Use in Heterogeneous Multi-Robot Exploration of Unknown Environments**
Longrui Yan†, Yiyu Wang†, ***Jingfan Tang***†, Yunpeng Lyu, Shizhe Zhao, Chao Cao, Zhongqiang Ren
In submission to IEEE Transactions on Robotics (T-RO), 2026

[2] **RPV*: Routing with Probabilistic Vertices and Its Use in Autonomous Exploration**
Jingfan Tang, Ruizhi Deng, Yunpeng Lyu, Shizhe Zhao, Zhongqiang Ren
International Symposium of Robotics Research (ISRR), 2026

[3] **EMOT*: Efficient Search Algorithm for the Multi-Objective Traveling Salesman Problem**
Jingfan Tang, Shizhe Zhao, Zhongqiang Ren
In submission to Artificial Intelligence (AIJ), 2026

[4] **Multi-robot Routing with Probabilistic Terminals for Cooperative Target Search**
Yunpeng Lyu, ***Jingfan Tang***, Zhongqiang Ren
In submission to IEEE International Conference on Robotics and Automation (ICRA), 2027

[5] **Goal2Pixel: Grounding Goals to Pixels for Vision-Language Navigation**
Muyi Bao, Yuxin Cai, Hang Xu, Zongtai Li, Jinxi He, ***Jingfan Tang***, Chen Lv, Ji Zhang, Yaqi Xie, Wenshan Wang
In submission to IEEE International Conference on Robotics and Automation (ICRA), 2027

Research Experience

Undergraduate Researcher, Shanghai Jiao Tong University Mar 2025 – Present

Project: Fast Search Algorithms for Multi-Objective Traveling Salesman Problems

- Addressed the multi-objective TSP, which seeks all Pareto-optimal paths under multiple conflicting objectives, and developed **EMOT***, an exact search framework with guarantee to find all cost-unique Pareto-optimal paths.
- Introduced state-based Pareto frontiers keyed on the last vertex and visited set, so that partial paths in one frontier share the same heuristic and are compared by cost vectors only, and designed a novel pruning rule using convex combinations of objective costs.
- Demonstrated **over an order-of-magnitude** speedup over baselines with vertex-based frontiers and **3× to 10,000×** faster performance than existing exact integer programming and dynamic programming methods on bi- and tri-objective TSP instances.

Project: Adaptive Autonomous Exploration System with Map Prediction

- Addressed exploration with learned map prediction, which is uncertain and suffers from train-test distribution shift, by formulating **Routing with Probabilistic Vertices (RPV)** that models prediction uncertainty as per-vertex existence probabilities.
- Developed **RPV***, an optimal A*-like solver with dominance pruning and admissible heuristics, and built an adaptive exploration system that fine-tunes the map predictor online from self-labeled observations within a single run.
- Reduced exploration time and travel distance by **13–21%** over state-of-the-art methods in simulated office and forest environments, and deployed the system on an AgileX Scout Mini robot in indoor and outdoor environments.

Project: Hierarchical Multi-Robot Exploration System for Heterogeneous Agents

- Addressed exploration with heterogeneous robots that can reach different terrain by formulating it as min-max

multi-robot routing with assignment constraints, and developed **RCH**, a bounded suboptimal search algorithm, within the **HEHA** exploration system.

- Designed RCH as an A*-like search over joint partial paths that expands only the robot with the least cost, making the branching factor independent of the number of robots, together with dominance pruning, an MST-based heuristic, and anytime focal search.
- Reduced makespan by **47%** over routing baselines and exploration time by up to **30%** over existing exploration methods, and deployed HEHA on a Unitree Go2 and an AgileX Scout Mini in a real environment with stairs.

Advisor: Prof. Zhongqiang Ren

Research Intern, Carnegie Mellon University Jan 2026 – Aug 2026

Project: Agentic Task Planning with Unrealizable Sub-Instructions (Ongoing)

- Studying long-horizon task planning where an LLM decomposes a language instruction into sub-instructions, some of which are **unrealizable** with the robot's skills or the current environment.
- Developing an agentic framework that reasons over a skill library to detect unrealizable sub-instructions before execution and triggers fallback mechanisms to revise or replace them, instead of executing infeasible plans.

Project: Language-Instructed Object Navigation Benchmark

- Proposed a benchmark for long-horizon embodied navigation that couples coarse-grained language instruction following with downstream object-goal navigation.
- Constructed staged navigation tasks from landmark-based trajectories and sub-instructions with difficulty control based on semantic and visibility cues, and built rollout and evaluation pipelines for language grounding and object-search efficiency.

Advisor: Prof. Ji Zhang

Skills

- **Programming Languages**: C, C++, Python, Matlab
- **Framework and Operating System**: Linux, Git, ROS (Robot Operating System)
- **Languages**: Mandarin, English

Honor and Awards

- **Dean's List**, Carnegie Mellon University, 2026
- **The John Wu & Jane Sun Merit Scholarship (Top 2% in overall performance)**, Shanghai Jiao Tong University, 2025
- **Undergraduate Excellent Scholarship (Top 5% in academic performance)**, Shanghai Jiao Tong University, 2024–2025
- **The Yu Liming Scholarship (Top 5% in overall performance)**, Shanghai Jiao Tong University, 2024

Services

- **Teaching Assistant of Course MATH2030J Discrete Mathematics**
  Lecturer: Dr. Runze Cai.
- **Teaching Assistant of Course MATH2550J Honor Multivariate Calculus**
  Lecturer: Dr. Runze Cai.
- **Teaching Assistant of Course ECE2810J Data Structure and Algorithms**
  Lecturer: Dr. Weikang Qian.
- **Organizer of CMU-VLN-Challenge 2026**
- **Reviewer for ICRA 2027, IROS 2026, Autonomous Robots 2026**
- **Leader and Advisor of GC Student Advising Center**
  Conducted Workshops on graduate school application (2025 Summer), and on freshman course selection (2024 Fall, 2025 Fall).